

Reversal-Space Tomography of Weak Altermagnetic Exchange in MnF_2 : Path-Validated Signed Readout Engineering

John Haines^{1,*}

¹*Independent Researcher*

(Dated: September 12, 2026)

Weak symmetry-selective interactions need not be easiest to identify in the largest spectral feature they influence. In MnF_2 , a seventh-neighbor exchange imbalance on the few- μeV scale coexists with a much larger dipolar contribution to the observed magnon doublet, while polarized-neutron and longitudinal-field protocols access different signed consequences of the same symmetry-breaking environment. We formulate these controls as *reversal-space tomography*: a quantitative experimental-design framework in which partner momentum, field sign, incident polarization, and magnetic-domain sign are treated as binary design coordinates and the measured response is decomposed into exact Walsh–Hadamard factorial contrasts. Interpreting those contrasts as characters of a physical $(\mathbb{Z}_2)^N$ reversal group requires an additional path-consistency gate: the implemented reversals must be involutive and mutually commuting on the prepared state, within experimental sensitivity. We introduce explicit involution and reversal-loop residuals for testing that assumption. We fix the microscopic convention $\delta J_7 \equiv J_{7b} - J_{7a}$ and separate it from the model-specific magnon detuning $\Lambda_{\mathbf{Q}} = \alpha_{\mathbf{Q}} \delta J_7$. For a minimal two-mode field model, $\Omega^2(q, b, d) = D^2 + [qd \alpha_{\mathbf{Q}} \delta J_7 + bH_0]^2$, the signed exchange appears linearly in the qbd contrast while the common dipolar baseline is even. We derive local readout gains and quantify how longitudinal field, compensation-field tracking, domain bias, and polarization analysis restore signed sensitivity. We also show explicitly that a second mechanism with the same qd transformation law contaminates the same qbd channel, so source specificity is conditional on the microscopic model or independent constraints. A Fisher-information form provides a common metric for comparing acquisition designs. The framework does not amplify the microscopic coupling; it makes weak terms identifiable only to the extent that reversal paths close, same-parity competitors are controlled, and predicted nulls survive.

I. INTRODUCTION

Altermagnets combine compensated collinear magnetic order with crystal-symmetry-enforced momentum-dependent spin structure distinct from conventional ferromagnets and antiferromagnets.[1, 2] MnF_2 has become a particularly useful test case because several physically distinct fine-structure scales coexist in a simple rutile antiferromagnet. Unpolarized inelastic neutron scattering constrained a resolvable altermagnetic magnon splitting,[5] polarized-neutron measurements subsequently resolved a dipolar-dominated doublet together with a domain-dependent polarization-odd chiral response,[6] and a recent first-principles study proposed a longitudinal-field protocol that converts a weak seventh-neighbor exchange imbalance into signed field-dependent observables.[7]

These results define an experimental inverse problem. The recorded spectrum contains contributions that differ greatly in magnitude and that transform differently under partner momentum, magnetic field, neutron polarization, domain reversal, and strain. A fit to one spectrum at one experimental setting can reproduce a line shape without uniquely identifying the microscopic term responsible for a weak feature. Conversely, a deliberately chosen reversal can suppress a large symmetry-even background while retaining a small symmetry-odd contribution.

The present work develops that observation into *reversal-space tomography*. The name refers to reconstruction of transformation-parity coefficients from controlled measurements, not to quantum-state tomography. Every scalar response measured on a complete binary factorial design has an exact Walsh–Hadamard expansion as a function of the *labels* assigned to the acquired vertices. A stronger statement—that those coefficients are characters of a physical $(\mathbb{Z}_2)^N$ reversal group—requires the implemented controls to act as independent involutions on a common prepared state. Field hysteresis, domain retraining, strain relaxation, or preparation memory can violate that condition. We therefore separate an always-valid factorial decomposition from a path-validated group interpretation. The transform does not manufacture information or guarantee a multiplex signal-to-noise advantage; it separates measured parity while exposing same-parity mechanisms and path dependence as genuine identification problems.

This paper is the readout-engineering layer of a broader source-to-observable program. The source-side coordinate is fixed here using the convention of Jiang *et al.*,

$$\boxed{\delta J_7 \equiv J_{7b} - J_{7a}}, \quad (1)$$

with the corresponding bond geometry stated in Ref. 7. This convention matters because the a/b labels used in different MnF_2 parameterizations are not interchangeable without relabeling. The measurement-side problem is kept distinct: neutron polarization, domains, finite resolution, and acceptance determine whether a generated response

* Haines.jj@gmail.com; ORCID: 0009-0001-5809-4690

reaches the recorded observable. The purpose of Paper III is neither to redefine the microscopic source nor to rederive the polarized-neutron cross section, but to ask a quantitative design question: *which controlled reversals provide the most informative signed readout of the weak source, and what nulls must accompany that assignment?*

II. SOURCE-TO-READOUT HANDOFF AND SIGN DISCIPLINE

The microscopic coefficient in Eq. (1) is not itself the measured magnon splitting. For a specified momentum and magnetic domain, write the exchange contribution to the two-mode detuning as

$$\Lambda_{\mathbf{Q}} = \alpha_{\mathbf{Q}} \delta J_7, \quad (2)$$

where $\alpha_{\mathbf{Q}}$ contains the model-specific spin-wave mapping from the bond-level coefficient to the relevant magnon sector. Partner momentum and time-reversed domain can reverse the effective detuning without changing the scalar convention chosen for δJ_7 . We therefore write

$$\Lambda(q, d) = qd \alpha_{\mathbf{Q}} \delta J_7, \quad q, d \in \{-1, +1\}. \quad (3)$$

This notation prevents three conceptually different signs from being conflated: the bond-label convention defining δJ_7 , the momentum character q , and the magnetic-domain character d .

At generic zero field, dipolar mixing D and the exchange detuning enter the minimal doublet separation in quadrature,

$$\Omega_0 = \sqrt{D^2 + \Lambda^2}. \quad (4)$$

The observable is therefore even under $\Lambda \rightarrow -\Lambda$ and cannot determine the microscopic sign. Jiang *et al.* explicitly identify this sign loss and propose longitudinal field as a signed readout.[7] Their supplemental calculation also identifies a momentum line on which the relevant projected dipolar mixing can vanish, leaving an exchange-driven zero-field splitting; this provides a useful magnitude cross-check but remains sign blind at zero field.[7]

The signed-readout problem is thus not equivalent to maximizing the total splitting. It is a problem of selecting an observable with a nonzero derivative with respect to the target coordinate and a transformation pattern that rejects competing terms.

III. FACTORIAL DECOMPOSITION AND THE PATH-CONSISTENCY GATE

Let the experiment have N binary controls

$$\mathbf{x} = (x_1, \dots, x_N), \quad x_i \in \{-1, +1\}, \quad (5)$$

and let $y(\mathbf{x})$ be a scalar estimator extracted from the data, such as a fitted mode separation, squared splitting,

integrated intensity, or polarization asymmetry. For every subset $A \subseteq \{1, \dots, N\}$ define the character

$$\chi_A(\mathbf{x}) = \prod_{i \in A} x_i, \quad \chi_{\emptyset} = 1. \quad (6)$$

The characters obey

$$\sum_{\mathbf{x}} \chi_A(\mathbf{x}) \chi_B(\mathbf{x}) = 2^N \delta_{AB}, \quad (7)$$

so every function on the 2^N vertices has the exact expansion

$$y(\mathbf{x}) = \sum_A C_A \chi_A(\mathbf{x}), \quad (8)$$

with

$$C_A = 2^{-N} \sum_{\mathbf{x}} \chi_A(\mathbf{x}) y(\mathbf{x}). \quad (9)$$

Equation (9) is a normalized Walsh–Hadamard transform after an ordering of vertices and basis functions is chosen. Hadamard methods have a long history in spectroscopy,[11] but here the transform is used as a parity-resolved factorial decomposition rather than as a universal multiplex advantage. Importantly, Eqs. (6)–(9) require only a complete labeled data cube; they do *not* by themselves prove that the laboratory controls realize a group action on one physical state.

For the experimentally natural controls partner momentum q , field sign b , and incident-polarization sign p , the basis is

$$\{1, q, b, p, qb, qp, bp, qbp\}. \quad (10)$$

Figure 1 shows the exact sign basis. A contribution odd in q and b but even in p resides in C_{qb} ; a polarization leakage term odd only in p resides in C_p . The transform separates *measured parity*. Assigning a microscopic origin to a coefficient remains a physical-model question, and interpreting the basis functions as physical group characters requires the path-consistency tests below.

A. When do factorial coordinates realize a reversal group?

Let r_i denote the laboratory operation intended to reverse control i on the full prepared state x , including latent variables such as domain population and preparation history. A physical $(\mathbb{Z}_2)^N$ interpretation requires, ideally,

$$r_i^2 x = x, \quad r_i r_j x = r_j r_i x \quad (11)$$

for all relevant i, j . These are stronger conditions than simply assigning ± 1 labels to endpoints. Because the full microscopic state is not directly observable, let $\mathbf{z}(x)$

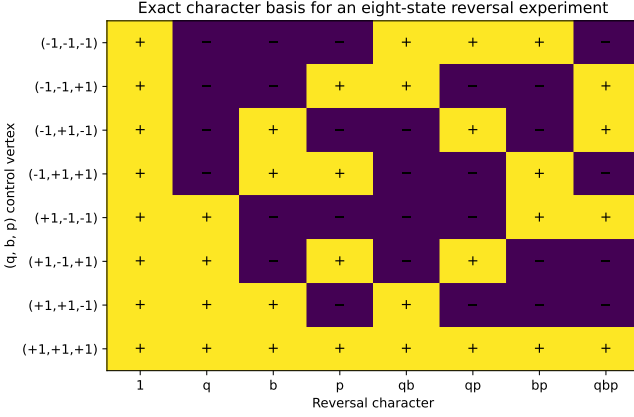


FIG. 1. Exact Walsh-Hadamard sign basis for an eight-state factorial experiment with partner momentum q , field sign b , and incident-polarization sign p . Algebraically, each column is a parity basis function on the labeled cube. It acquires the stronger interpretation of a $(\mathbb{Z}_2)^3$ group character only when the corresponding physical reversals pass the involution and path-commutation tests in the text. Microscopic assignment remains a separate model question. The schematic was generated with deterministic plotting code developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the definitions and displayed relationships were specified and verified by the author.

be a vector of experimentally monitored diagnostics—for example fitted mode positions and intensities, a domain-sensitive signal, polarization efficiency, sample position, and an independent strain or field monitor. Define the measured involution and loop residuals

$$\epsilon_i^{\text{inv}}(x) = \mathbf{z}(r_i^2 x) - \mathbf{z}(x), \quad (12)$$

$$\epsilon_{ij}^{\text{loop}}(x) = \mathbf{z}(r_i r_j x) - \mathbf{z}(r_j r_i x). \quad (13)$$

With residual covariance matrices Σ_i^{inv} and $\Sigma_{ij}^{\text{loop}}$, useful dimensionless diagnostics are

$$D_{i,\text{inv}}^2 = (\epsilon_i^{\text{inv}})^T (\Sigma_i^{\text{inv}})^{-1} \epsilon_i^{\text{inv}}, \quad (14)$$

$$D_{ij,\text{loop}}^2 = (\epsilon_{ij}^{\text{loop}})^T (\Sigma_{ij}^{\text{loop}})^{-1} \epsilon_{ij}^{\text{loop}}. \quad (15)$$

Small residuals in a sufficiently rich diagnostic set support, but do not logically prove, closure of the latent state. Significant residuals mean that the controls should be treated as path-dependent interventions; the Walsh-Hadamard coefficients remain valid *factorial contrasts*, but their interpretation as reversal-group characters must be downgraded or the state description enlarged to include history.

A concrete counterexample illustrates the distinction. Let the visible controls be field sign b and strain sign e , and let d be a hidden domain label. Suppose strain reversal is innocuous,

$$r_e(b, e, d) = (b, -e, d), \quad (16)$$

whereas field reversal retrains the domain only for $e = +1$,

$$r_b(b, e, d) = \begin{cases} (-b, e, d), & e = -1, \\ (-b, e, -d), & e = +1. \end{cases} \quad (17)$$

Starting from $(b, -1, d)$ gives

$$r_b r_e(b, -1, d) = (-b, +1, -d), \quad (18)$$

$$r_e r_b(b, -1, d) = (-b, +1, d). \quad (19)$$

The same visible endpoint $(b, e) = (-b, +1)$ therefore corresponds to different physical domain states depending on path. A single hypercube vertex is no longer a unique prepared state. Field-induced domain switching, strain-assisted retraining, thermal or optical annealing, and other hysteretic preparation effects are experimental realizations of this general failure mode.

If all 2^N vertices have equal independent variance σ^2 , then

$$\text{Var}(C_A) = \frac{\sigma^2}{2^N}. \quad (20)$$

At fixed total counting time, however, measuring more vertices also divides the available exposure. There is no generic free $\sqrt{2^N}$ sensitivity gain. The benefit is rejection of parity-incompatible backgrounds, direct tests of predicted null channels, and a transparent diagnosis of mechanisms that share the same parity.

IV. QUANTITATIVE LONGITUDINAL-FIELD TRANSDUCTION

Let $b = \pm 1$ denote the sign of a fixed longitudinal field magnitude B_0 , and define

$$H_0 = 2g_{\text{eff}}\mu_B B_0. \quad (21)$$

The minimal two-mode splitting can be written

$$\Omega^2(q, b, d) = D^2 + [qd\alpha_{\mathbf{Q}}\delta J_7 + bH_0]^2. \quad (22)$$

Expanding,

$$\Omega^2 = D^2 + (\alpha_{\mathbf{Q}}\delta J_7)^2 + H_0^2 + 2qbd\alpha_{\mathbf{Q}}\delta J_7 H_0. \quad (23)$$

For a complete $q \times b \times d$ design, the odd coefficient is therefore

$$C_{qbd} = 2\alpha_{\mathbf{Q}}\delta J_7 H_0. \quad (24)$$

The dipolar baseline and the squared magnitudes of exchange and field reside in the even sector. If d is fixed by a trained monodomain, the same term appears in C_{qb} with a domain-sign convention. If the beam averages two incoherent domains with $d = +1$ fraction ρ , a domain-odd coefficient is reduced by

$$\eta_D = 2\rho - 1, \quad (25)$$

so

$$C_{qb}^{\text{mix}} = 2\eta_D \alpha_{\mathbf{Q}} \delta J_7 H_0. \quad (26)$$

This is a measurement transduction factor, not a renormalization of the microscopic exchange.

The source assignment in Eq. (24) is conditional on the minimal Hamiltonian. To see the degeneracy explicitly, suppose a second microscopic coordinate ξ generates an indistinguishable qd -odd detuning with coefficient $\beta_{\mathbf{Q}}$,

$$\Lambda_{\text{tot}}(q, d) = qd(\alpha_{\mathbf{Q}} \delta J_7 + \beta_{\mathbf{Q}} \xi). \quad (27)$$

The same field construction then gives

$$C_{qbd} = 2H_0(\alpha_{\mathbf{Q}} \delta J_7 + \beta_{\mathbf{Q}} \xi). \quad (28)$$

Thus the qbd channel identifies a signed *source combination* unless competing same-parity coordinates are excluded by microscopic theory, constrained by another momentum dependence or probe, or separated through additional functional dependence. This is the explicit source-specificity limit of reversal-space tomography.

For a continuous field sweep, Eq. (22) gives the compensation field

$$B_{qd}^* = -\frac{qd \alpha_{\mathbf{Q}} \delta J_7}{2g_{\text{eff}} \mu_B}. \quad (29)$$

The two partner momenta therefore have equal-and-opposite compensation fields in the minimal model. Jiang *et al.* report $|B^*| \simeq 0.34$ T for their production parameterization and a fixed-field partner-splitting contrast of about 0.14 meV at 1 T; their resolution-convolved simulations retain these signatures for a 0.12 meV energy resolution.[7] Figure 2 illustrates the corresponding two-mode geometry using the representative exchange and dipolar scales reported in that work. It is not new neutron data.

The novelty of the reversal-space formulation is not the field protocol itself. It is the recognition that the signed field readout occupies a definite character sector and can therefore be combined with polarization and domain controls in a single overconstrained inference design.

V. LOCAL READOUT GAIN AND FISHER INFORMATION

A weak coefficient should not be judged by its fraction of the Hamiltonian norm alone. For a measured observable O_x under experimental condition x , define the local readout gain

$$\mathcal{G}_x = \left| \frac{\partial O_x}{\partial \delta J_7} \right|. \quad (30)$$

This quantity describes local transduction. It is not a universal amplification factor because it depends on the

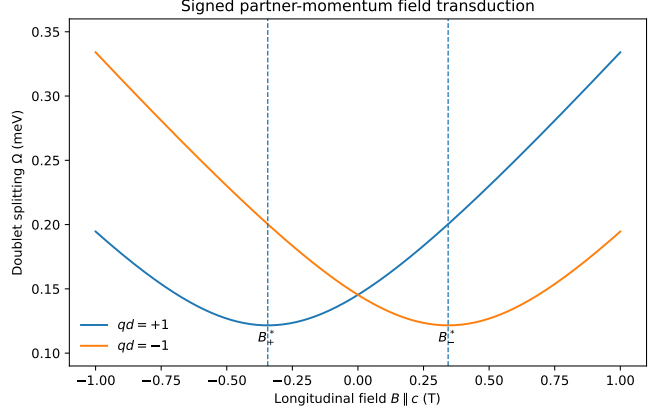


FIG. 2. Signed partner-momentum field transduction in the minimal two-mode model. Rotation-related effective detunings have opposite sign and therefore opposite compensation fields. The plotted scales use the representative exchange-only and dipolar contributions discussed by Jiang *et al.*[7] and reproduce $|B^*| \approx 0.34$ T and an approximately 0.14-meV partner contrast at 1 T for $g_{\text{eff}} = 2$. This is an analytic illustration, not experimental data. The deterministic plotting code was developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the equations and parameter use were specified and verified by the author.

observable, momentum, field, domain state, response kernel, and instrumental map.

For the generic zero-field splitting in Eq. (4), with $\Lambda = \alpha \delta J_7$,

$$\frac{\partial \Omega_0}{\partial \delta J_7} = \frac{\alpha^2 \delta J_7}{\sqrt{D^2 + (\alpha \delta J_7)^2}}. \quad (31)$$

Thus the local sensitivity to the signed source collapses linearly as $\delta J_7 \rightarrow 0$ when $D \neq 0$, and the observable remains even under sign reversal. In the dipolar-mixing-free limit $D \rightarrow 0$, $\Omega_0 = |\alpha \delta J_7|$ has a finite one-sided magnitude sensitivity $|\alpha|$, but still cannot identify sign.

Now consider the same-field partner contrast

$$\Delta \Omega(B) = \Omega(q = +1, B) - \Omega(q = -1, B) \quad (32)$$

for a fixed domain. Linearizing around $\delta J_7 = 0$ yields

$$\left. \frac{\partial \Delta \Omega}{\partial \delta J_7} \right|_0 = \frac{2\alpha H}{\sqrt{D^2 + H^2}}. \quad (33)$$

The field therefore converts a sign-blind quadratic dependence into a signed linear response. The normalized gain increases with H/D and saturates rather than diverging. Figure 3 contrasts the two cases. This is the precise sense in which readout engineering can make a small source more identifiable without making the microscopic coefficient larger.

For an acquisition yielding mean data vector $\boldsymbol{\mu}(\delta J_7)$ with covariance Σ , a natural local design metric is the Fisher information

$$\mathcal{I}_{\delta J_7} = \mathbf{g}^T \Sigma^{-1} \mathbf{g}, \quad \mathbf{g} = \frac{\partial \boldsymbol{\mu}}{\partial \delta J_7}. \quad (34)$$

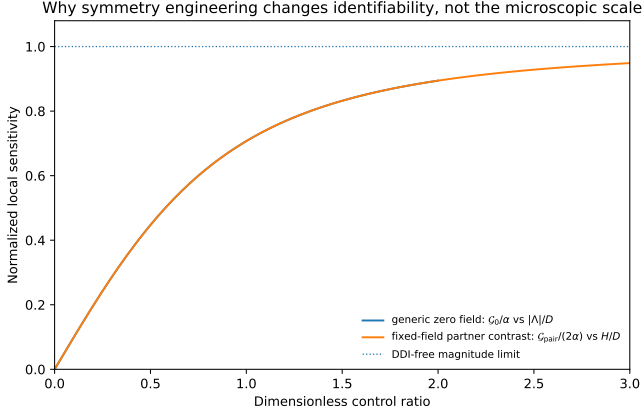


FIG. 3. Normalized local sensitivity for two idealized readouts. Generic zero-field splitting has vanishing signed sensitivity near the symmetry-restoring point when dipolar mixing is finite. The fixed-field partner contrast acquires finite first-order sensitivity controlled by H/D and saturates rather than diverging. The dotted limit denotes the dipolar-mixing-free magnitude sensitivity, which remains sign blind at zero field. Curves are analytic evaluations of Eqs. (31) and (33), not experimental fits. Deterministic plotting code was developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the equations and interpretation were specified and verified by the author.

For independent scalar estimators this reduces to

$$\mathcal{I}_{\delta J_7} = \sum_k \frac{1}{\sigma_k^2} \left(\frac{\partial \mu_k}{\partial \delta J_7} \right)^2. \quad (35)$$

Equation (34) separates two experimentally different objectives: increase physical transduction $|\partial \mu / \partial \delta J_7|$, or reduce estimator uncertainty and covariance. A reversal protocol can improve the second through common-mode rejection and overconstraint even when it does not increase the intrinsic derivative.

VI. POLARIZATION AND DOMAINS AS INDEPENDENT SIGNED CHANNELS

Incident-polarization reversal probes a different observable from the field-induced energy contrast. In the half-polarized neutron geometry, the polarization-odd signal isolates the antisymmetric dynamical correlation channel rather than merely measuring the same scalar spectral splitting with greater sensitivity.[4, 6] Near the symmetry-restoring point, a series-level leading-order form may be written schematically as

$$I_{\chi}^{\text{obs}}(\mathbf{Q}, \omega) = \mathcal{T}_R [\eta_D \mathcal{K}_{\chi}(\mathbf{Q}, \omega) \delta J_7 + O(\delta J_7^3)], \quad (36)$$

where $\mathcal{K}_{\chi}$ contains the mode eigenvectors, magnetic matrix elements, momentum and polarization geometry, dipolar mixing, form factors, and other model-specific factors, while $\mathcal{T}_R$ denotes instrumental transfer and resolution.

Equation (36) is not a fitted MnF_2 formula and does not imply that δJ_7 alone predicts the measured percentage asymmetry.

A trained domain and polarization reversal therefore supply an independent cross-validation of the source assignment. Agreement between a field-derived sign and a polarization/domain-derived sign under one microscopic model is stronger evidence than either magnitude alone because the two readouts occupy different experimental channels. Conversely, disagreement is scientifically useful: it signals an incorrect sign convention, an unmodeled interaction, domain mischaracterization, polarization leakage, or an incomplete response kernel.

Domain averaging is especially important. A rough 4:1 domain-population imbalance corresponds to $\eta_D \approx 0.6$ as an illustrative central value; the actual measured population and its uncertainty should be propagated in any quantitative fit. A monodomain sample increases a purely domain-odd macroscopic response by removing this dilution but does not change δJ_7 itself. Equal domain populations force every domain-odd macroscopic coefficient to zero even when each domain has a finite local response.

VII. A COMMON READOUT MATRIX

Table I summarizes the principal MnF_2 readouts in one language. The ranking is by identifiability of the signed weak source, not by absolute signal magnitude.

The table suggests a practical hierarchy. A longitudinal-field sweep of both partner momenta is the cleanest proposed signed spectroscopic readout because the opposite compensation fields provide both magnitude and sign within the stated model. A fixed-field partner contrast is a fast sign cross-check. Polarized-neutron chirality supplies an independent tensor-channel validation once domains and polarization efficiency are characterized. A dipolar-mixing-free zero-field momentum provides a useful magnitude cross-check. Strain becomes a microscopic δJ_7 control only after its coupling to the exchange imbalance is independently calibrated.

VIII. STRAIN AS A CALIBRATED EXTENSION

Strain is experimentally compelling because rutile altermagnets possess symmetry-allowed piezomagnetic and altermagnetic strain responses.[3, 8] Recent elastocaloric measurements on MnF_2 combine strain and magnetic field to provide a bulk thermodynamic probe of multipolar/altermagnetic order.[9] These results establish strain as a physically meaningful control coordinate, but they do not by themselves determine how the microscopic seventh-neighbor imbalance changes with a particular strain component.

TABLE I. Readout-engineering matrix for weak altermagnetic exchange in MnF_2 . “Signed” means that the observable can distinguish the sign of the effective source under an explicitly fixed convention.

Readout	Controls	Leading relation	Sign?	Principal caveat
Generic zero-field doublet	q	$\Omega_0 = \sqrt{D^2 + (\alpha\delta J_7)^2}$	No	quadratic sign loss
DDI-free zero-field window	q	$\Omega_0 \rightarrow \alpha\delta J_7 $	No	magnitude only
Compensation field	q, B	$B_{qd}^* = -qd\alpha\delta J_7 / (2g\mu_B)$	Yes	model and field calibration
Fixed-field partner contrast	q, b	$C_{qb}^{\text{mix}} = 2\eta_D\alpha\delta J_7 H_0$	Yes	domain dilution, fit covariance
Polarization-odd intensity	p, d	$I_{\chi}^{\text{obs}} = \mathcal{T}_R[\eta_D\mathcal{K}_{\chi}\delta J_7 + \dots]$	Conditional	response kernel, polarization, domains
Strain response	$e + \text{reversals}$	$\partial_e O = \kappa_7 \partial_{\delta J_7} O + (\partial_e O)_{\text{direct}}$	Not alone	microscopic strain calibration

Write, locally,

$$\delta J_7(\epsilon) = \delta J_7(0) + \kappa_7 \epsilon + O(\epsilon^2), \quad \kappa_7 = \frac{\partial \delta J_7}{\partial \epsilon}. \quad (37)$$

For any measured observable O ,

$$\left. \frac{\partial O}{\partial \epsilon} = \kappa_7 \frac{\partial O}{\partial \delta J_7} + \frac{\partial O}{\partial \epsilon} \right|_{\text{direct}}. \quad (38)$$

The second term includes direct strain dependence of dipolar, anisotropy, g -factor, structural, and other exchange contributions. Therefore a nonzero strain-odd parity coefficient is evidence that strain changes the measured response with a specified transformation character; it is not, by itself, a measurement of κ_7 .

This distinction provides a concrete calibration program. First determine the strain-parity coefficients experimentally. Then use first-principles or independent spectroscopy to decompose the direct terms in Eq. (38). Only after that step should strain be interpreted as a calibrated knob for tuning δJ_7 itself.

IX. ACQUISITION, COVARIANCE, AND FALSIFICATION

The factorial algebra requires comparable labeled vertices, while the group-character interpretation additionally requires path consistency. Real neutron experiments are vulnerable to source drift, detector changes, polarization-efficiency drift, sample motion, hysteresis, domain retraining, strain relaxation, and field-history effects. Frequent state changes can reduce bias in two-state neutron measurements, and acquisition order should therefore be considered part of the estimator design.[10]

A hardened implementation therefore contains a path audit before the symmetry interpretation is accepted. Prepare and independently characterize a common baseline state; for each intended reversal r_i , acquire $x \rightarrow r_i x \rightarrow r_i^2 x$ and evaluate Eq. (12); for each important pair (i, j) , acquire both orderings $r_i r_j$ and $r_j r_i$ from matched baseline histories and evaluate Eq. (13); include domain-sensitive and control-monitor observables in z rather than using the target spectrum alone. Only after those checks should the data be interpreted as characters of $(\mathbb{Z}_2)^N$. If the gate fails, either enlarge the state label to include history or retain the coefficients strictly as factorial contrasts.

After this audit, choose a symmetry-related momentum pair; acquire balanced reversal blocks in randomized or pseudorandom order; repeat reference vertices to diagnose drift; fit all spectra globally with common physical parameters and resolution convolution; transform both fitted observables and appropriate residual summaries into parity coefficients; and propagate the full covariance. Optical and field/stress routes to domain manipulation have been demonstrated in MnF_2 ,[12] making domain state a particularly important latent variable to monitor.

For coefficient vector $\mathbf{C} = 2^{-N} H_N \mathbf{y}$, where H_N is the unnormalized Hadamard matrix,

$$\Sigma_C = 2^{-2N} H_N \Sigma_y H_N^T. \quad (39)$$

Shared normalization, common fitted parameters, and repeated-vertex correlations should remain in Σ_y rather than being discarded before the transform.

The framework is useful only if it can fail. The central falsification tests are:

1. **Involution test.** For each claimed binary reversal, ϵ_i^{inv} in Eq. (12) must be statistically compatible with closure in the chosen diagnostic set. Failure means that the operation is hysteretic or history changing rather than an involution.
2. **Reversal-loop test.** For each important pair of controls, $\epsilon_{ij}^{\text{loop}}$ in Eq. (13) must be compatible with zero under matched histories. Failure means that the two controls cannot be interpreted as commuting generators of $(\mathbb{Z}_2)^N$ without enlarging the state description.
3. **Exchange-null test.** If $\delta J_7 = 0$ in the minimal source model, its contribution to the qbd field coefficient must vanish while a finite dipolar splitting can remain. A nonzero coefficient then implies another mechanism in the same parity sector.
4. **Same-parity source test.** Introduce or calculate an allowed competitor such as ξ in Eq. (27). If the available observables depend only on $\alpha\delta J_7 + \beta\xi$, the experiment identifies that combination, not δJ_7 uniquely. A second momentum, probe, or independent microscopic constraint is then required.
5. **Field-null test.** At $B = 0$, the signed field-exchange cross term vanishes. A zero-field splitting alone cannot determine the sign of δJ_7 .

6. **Domain-balance test.** At $\rho = 1/2$, domain-odd macroscopic coefficients must vanish in the incoherent two-domain model.
7. **Polarization-null test.** The polarization-odd chiral coefficient must vanish as the effective incident polarization tends to zero, while polarization-even magnetic intensity can remain finite.
8. **Partner-reversal test.** Within the minimal field model, the partner relation $\Omega_+(B) = \Omega_-(-B)$ must hold after sign conventions are fixed. Reproducible violation indicates additional partner-odd physics or an inadequate truncation.
9. **Even-channel leakage test.** A known symmetry-even calibration feature should not appear strongly in a nominally odd factorial coefficient. If it does, acquisition or normalization requires revision.
10. **Strain-identification test.** A strain-odd signal does not identify κ_7 unless the direct term in Eq. (38) is constrained independently.

The objective is therefore not merely to obtain a nonzero difference. A credible source assignment should reproduce the complete pattern of predicted signs, zeros, partner relations, and cross-channel consistencies.

X. DISCUSSION

Reversal-space tomography separates three operations that are easily conflated. First, microscopic theory identifies a source coordinate such as δJ_7 . Second, the magnetic response maps that coordinate into mode energies, eigenvectors, and correlation tensors. Third, an experiment projects those responses through field, polarization, domains, resolution, and analysis choices. A small coefficient can be the first allowed source in a selected channel without being large in the Hamiltonian norm, and a large physical response can still be unidentifiable if the recorded observable is even in the source or averages over opposite domains.

For MnF_2 , longitudinal field is attractive because it creates a linear cross term between the weak exchange detuning and a known external control. The compensation-field position is a particularly direct signed coordinate within the minimal model. Polarization is complementary rather than redundant: it measures an antisymmetric dynamical correlation channel, not the same energy contrast. Domain preparation acts as both a sign coordinate and a transmission factor. The strongest experiment is therefore not necessarily the setting with the largest raw spectral separation; it is the design that gives a large derivative with respect to the target source, suppresses parity-incompatible backgrounds, and contains enough independent reversals to test the source assignment.

The framework states two distinct identifiability limits. First, if two microscopic mechanisms transform identically

under every available reversal and enter the same response tensor with indistinguishable functional dependence, no Walsh–Hadamard projection can separate them; Eq. (28) is the explicit counterexample. Second, even perfectly separated factorial coefficients do not become symmetry characters if the physical controls fail the path-consistency gate. In the first case another momentum region, probe, or microscopic prior is required. In the second, history must be modeled as part of the state or the analysis must remain at the level of factorial contrasts. Neither limitation is a failure of the algebra; each certifies what the current experiment has not identified.

The present work therefore does not claim a universal hiddenness ratio or a universal inverse-amplification law. Nor does it claim experimental validation of the longitudinal-field prediction, which remains a proposed readout based on the cited first-principles model. Its contribution is a common quantitative language for deciding which measurements would make that prediction identifiable and falsifiable.

XI. CONCLUSION

We have developed reversal-space tomography as a quantitative readout-engineering framework for weak antiferromagnetic exchange in MnF_2 . The microscopic source convention $\delta J_7 = J_{7b} - J_{7a}$ is kept distinct from the momentum- and domain-dependent magnon detuning it generates. Any complete binary design admits an exact Walsh–Hadamard decomposition into orthogonal factorial parity coefficients. Those coefficients acquire the stronger interpretation of $(\mathbb{Z}_2)^N$ reversal-group characters only after the implemented controls pass involution and reversal-loop tests on a common prepared state. In the minimal longitudinal-field model, the δJ_7 contribution occupies the qbd coefficient of the squared splitting,

$$C_{qbd} = 2\alpha_Q \delta J_7 H_0, \quad (40)$$

while dipolar and squared-magnitude terms remain even. A same-parity competitor contributes to the same coefficient, so source specificity is conditional on the microscopic model or independent constraints. Domain mixing reduces domain-odd channels by η_D , polarization reversal provides an independent chiral response channel, and strain is a calibrated extension whose direct magnetoelastic contributions must be separated from $\partial \delta J_7 / \partial \epsilon$.

Local readout gain and Fisher information make the design principle quantitative. Generic zero-field splitting loses signed sensitivity near the symmetry-restoring point; longitudinal field restores a first-order signed derivative; a dipolar-mixing-free momentum can improve magnitude sensitivity without restoring sign; and a balanced reversal design converts null channels into direct diagnostics of contamination and model incompleteness. The resulting experimental prescription is simple: identify a weak interaction by the complete way its observables transform, not merely by the size of one spectral feature.

ACKNOWLEDGMENTS

No external funding was used for this theoretical and methodological analysis. The author acknowledges the experimental and theoretical groups whose published and preprint work on MnF_2 provides the quantitative basis for the proposed protocol. OpenAI ChatGPT (GPT-5.6 Sol) was used substantively during manuscript preparation to assist with literature organization, derivation and sign-convention cross-checking, manuscript restructuring, language editing, and development of deterministic plotting code for schematic figures. The author directed the scientific scope and claims and verified AI-assisted outputs against the cited primary literature, explicit algebraic derivations, and the final plotted relationships. The author retains full responsibility for all claims, equations, citations, figures, and conclusions.

DATA AVAILABILITY

This theoretical and methodological study reports no new experimental data and does not independently refit raw neutron-scattering data. Numerical values used for analytic illustrations are taken from the cited public literature. The source equations and deterministic plotting code used for the schematic figures are included in the accompanying manuscript package.

CONFLICT OF INTEREST

The author declares no competing financial interest related to this manuscript.

-
- [1] L. Šmejkal, J. Sinova, and T. Jungwirth, “Beyond Conventional Ferromagnetism and Antiferromagnetism: A Phase with Nonrelativistic Spin and Crystal Rotation Symmetry,” *Phys. Rev. X* **12**, 031042 (2022), doi:10.1103/PhysRevX.12.031042.
 - [2] P. A. McClarty and J. G. Rau, “Landau Theory of Altermagnetism,” *Phys. Rev. Lett.* **132**, 176702 (2024), doi:10.1103/PhysRevLett.132.176702.
 - [3] S. Bhowal and N. A. Spaldin, “Ferroically Ordered Magnetic Octupoles in d -Wave Altermagnets,” *Phys. Rev. X* **14**, 011019 (2024), doi:10.1103/PhysRevX.14.011019.
 - [4] P. A. McClarty, A. Gukasov, and J. G. Rau, “Observing Altermagnetism Using Polarized Neutrons,” *Phys. Rev. B* **111**, L060405 (2025), doi:10.1103/PhysRevB.111.L060405.
 - [5] V. C. Morano, Z. Maesen, S. E. Nikitin, J. Lass, D. G. Mazzone, and O. Zaharko, “Absence of Altermagnetic Magnon Band Splitting in MnF_2 ,” *Phys. Rev. Lett.* **134**, 226702 (2025), doi:10.1103/PhysRevLett.134.226702.
 - [6] Q. Faure, D. Bounoua, V. Balédent, A. Gukasov, V. O. Garlea, A. Ribeiro, J. G. Rau, S. Petit, and P. McClarty, “Altermagnetism Revealed by Polarized Neutrons in MnF_2 ,” to appear in *Science*; arXiv:2509.07087 (2025), doi:10.48550/arXiv.2509.07087.
 - [7] G. Jiang, F. Wang, Y. Wang, F. Zheng, B. Xi, H.-G. Luo, and J. Zhao, “Field-amplified readouts of weak altermagnetic exchange in MnF_2 ,” arXiv:2608.26540v2 (2026), doi:10.48550/arXiv.2608.26540.
 - [8] M. Khodas, S. Mu, I. I. Mazin, and K. D. Belashchenko, “Tuning of altermagnetism by strain,” *Phys. Rev. B* **113**, 104422 (2026), doi:10.1103/r7mn-yklk.
 - [9] R. Ohlendorf, L. Buiarelli, H. M. L. Noad, A. P. Mackenzie, R. M. Fernandes, T. Birol, J. Schmalian, and E. Gati, “Probing Multipolar Order in the Candidate Altermagnet MnF_2 through the Elastocaloric Effect under Strain,” *Phys. Rev. Lett.* **137**, 056702 (2026), doi:10.1103/svrz-315w.
 - [10] M. R. Fitzsimmons and T. R. Charlton, *Nucl. Instrum. Methods Phys. Res. A* **941**, 162330 (2019), doi:10.1016/j.nima.2019.06.071.
 - [11] E. D. Nelson and M. L. Fredman, *J. Opt. Soc. Am.* **60**, 1664–1669 (1970), doi:10.1364/JOSA.60.001664.
 - [12] T. Higuchi and M. Kuwata-Gonokami, “Control of antiferromagnetic domain distribution via polarization-dependent optical annealing,” *Nat. Commun.* **7**, 10720 (2016), doi:10.1038/ncomms10720.